

Assignment 4: Predicting PM2.5 Air Quality Concentrations from AirQo Sensor Data

AI Deployment & Scalability

Deadline: 4th October 2026, 24:59 (Kampala time)

Late submissions are not accepted unless prior arrangement has been made. Read this entire brief before you start -several stages depend on decisions you make earlier in the pipeline.

Total Marks: 20%

1. About AirQo

AirQo is an air quality monitoring initiative based at Makerere University, Uganda, founded by Engineer Bainomugisha and recognized as a Google.org "Leader to Watch" in 2022. It designs and locally manufactures low-cost air quality monitors -the AirQo Binos Monitor -engineered specifically for the African context: solar- or mains-powered, built to withstand dust and heat, and able to transmit data over 2G networks where connectivity is unreliable. Beyond fixed installations, AirQo also mounts sensors on motorcycle taxis ("boda-bodas") to extend spatial coverage.

The network now spans over 350 monitors across 8+ major African cities, including Kampala, Nairobi, Lagos, and Accra, measuring PM2.5, PM10, and NO2. Readings are made available through a mobile app, a real-time data platform, and a public API, and are increasingly used by researchers, city authorities, and public health decision-makers to inform environmental policy.

This assignment works with a slice of that sensor network: hourly PM2.5 and PM10 readings from 149 sensors across 142 zones.

2. Understanding PM2.5 and PM10

What they are. PM2.5 and PM10 are particulate matter -mixtures of solid particles and liquid droplets suspended in air -classified by diameter. PM10 refers to particles up to 10 micrometres across (about one-seventh the width of a human hair); PM2.5 refers to the finer fraction, particles up to 2.5 micrometres, roughly 30 times smaller than the width of a human hair.

Sources. Common sources across Ugandan cities include vehicle exhaust (especially older, poorly maintained fleets and traffic congestion), burning of biomass and charcoal for cooking, open burning of waste, unpaved roads and construction dust, and small-scale industrial and generator emissions. PM10 tends to come more from mechanical processes -road dust, construction, wind-blown soil - while PM2.5 includes both directly emitted ("primary") particles from combustion and particles that form in the atmosphere from gaseous precursors ("secondary" particles).

Why PM2.5 is especially dangerous. Because of its small size, PM2.5 bypasses the body's natural filtering (nose and upper airways) and penetrates deep into the lungs, where it can cross into the bloodstream. Long-term exposure is linked to respiratory disease, cardiovascular disease, stroke,

and premature death; children, the elderly, and people with existing respiratory or heart conditions are most vulnerable. PM₁₀, while still harmful, mostly deposits in the upper respiratory tract and is generally considered less dangerous at equivalent concentrations.

Influence of temperature and humidity. Meteorology has a measurable effect on PM_{2.5} concentrations: temperature affects the height of the atmospheric boundary layer and the rate of photochemical reactions that form secondary aerosols, while humidity promotes hygroscopic growth of particles and can influence both particle formation and how readily they settle out of the air. Low wind speed and temperature inversions (common in the early morning and evening) trap pollutants near the ground and tend to coincide with PM_{2.5} peaks. PM₁₀, being dominated by mechanically generated dust rather than atmospheric chemistry, is far less sensitive to temperature and humidity -this is why the two pollutants should not be modelled identically.

3. Air Quality and Public Health in Uganda

Kampala is consistently ranked among the more polluted major cities globally. A four-year continuous monitoring study (2018–2021) using a reference-grade instrument found an annual average PM_{2.5} concentration of 39 µg/m³ for the city -roughly eight times the WHO annual guideline of 5 µg/m³ -with concentrations peaking during the dry seasons and during morning/evening traffic hours. Over that same period, an estimated 7,257 deaths in Kampala were attributable to long-term PM_{2.5} exposure above WHO guideline levels. Pollution burdens are not evenly distributed: informal settlements and "ghetto" zones, often located near busy roads, waste-burning sites, or dense low-ventilation housing with widespread biomass cooking, typically experience some of the highest exposure levels in the city.

This is why continuous tracking -and, further, forecasting -of PM_{2.5} concentrations matters well beyond academic interest:

- **Health sector.** Early warning of poor air quality days allows hospitals and clinics to anticipate spikes in respiratory and cardiovascular admissions, and allows the public (especially asthma patients, the elderly, and caregivers of young children) to take precautions.
- **National Environment Management Authority (NEMA).** Reliable, granular concentration data supports enforcement of air quality standards, siting decisions, and evidence-based environmental regulation.
- **Ministry of Energy.** Household air pollution is closely tied to cooking fuel choices (charcoal, biomass, LPG); trend data helps target clean-energy transition programmes where they matter most.
- **Ministry of Works and Transport.** Since vehicle emissions and traffic congestion are major PM_{2.5} contributors, forecasted pollution levels can inform traffic management, fleet-age regulation, and road infrastructure planning.

Predictive models -the kind you will build in this assignment -turn a sparse, sensor-limited picture into forward-looking, city-wide situational awareness that these institutions can act on.

4. Air Quality Index (AQI) Categories

Use the following PM2.5 concentration bands throughout this assignment (EDA visualizations, model evaluation, and forecast interpretation all reference these categories):

PM2.5 Concentration ($\mu\text{g}/\text{m}^3$)	Category
0 – 9	Good
9.101 – 35.49	Moderate
35.491 – 55.49	Unhealthy for Sensitive Groups
55.491 – 125.49	Unhealthy
125.491 – 225.49	Very Unhealthy
225.491 – ∞	Hazardous

5. Dataset Description

You are provided a dataset of **14,600 hourly records** collected from **149 sensors across 142 zones**, covering **January to July 2026**.

Field	Description
site_name	Name of the monitoring site
sensor_name	Identifier for the physical sensor unit
sample_rate	Measurement frequency (hourly)
network_note	Notes on device/network connectivity status
latitude, longitude	Geographic coordinates of the site
temp	Ambient temperature reading
humidity	Ambient relative humidity reading

Field	Description
pm2_5_calibrated_value	Calibrated PM2.5 concentration ($\mu\text{g}/\text{m}^3$)
pm10_calibrated_value	Calibrated PM10 concentration ($\mu\text{g}/\text{m}^3$)
site_id	Unique identifier for the monitoring site
device_name	Name/identifier of the physical device

Note: for temp, humidity, pm2_5_calibrated_value, pm10_calibrated_value, latitude, and longitude, a value of 0 or an empty field should be treated as a **null**, not a genuine zero reading. Some devices will also go offline for stretches of time -this is expected in real-world low-cost sensor networks, and your data pipeline and modelling architecture should be designed to accommodate it rather than treat it as an anomaly to eliminate.

6. Assignment Objective

Ultimately, this assignment asks you to build a complete, deployable pipeline that **predicts PM2.5 concentration** from the AirQo sensor data (in zipped file) -from raw data to a trained, evaluated, monitored, and cloud-deployed model.

7. Methodology

Work through the following stages in order. Each stage's output feeds the next, so do not skip validation steps.

7.1 Data Preparation

- Clean the dataset so that it is consistent throughout: resolve the null cases defined in dataset, and use a statistically justified approach to handle them (do not simply drop every incomplete row without reasoning about what you lose). Consult at least one or two published papers on PM2.5 sensor data preparation to understand accepted practice for when to impute, interpolate, or drop records -cite what you use. (Check AirQo pp)
- Decide, with justification, which fields are actually useful predictors -not every column in the dataset needs to enter the model.
- Resample the data so that each sensor has a complete hourly time index across the full January–July 2026 window (including hours where the device was offline, represented as missing rather than absent).
- Enforce consistent data types across every field (timestamps, numeric fields, identifiers, etc.).
- Carry out any other cleaning/preparation steps you judge necessary, and document your reasoning.
- Remember: temp and humidity are relevant predictors for PM2.5 but **not** for PM10 (Section 2) -let that shape your feature selection.

7.2 Exploratory Data Analysis

Select roughly **5 sensors located within Kampala** for this section.

- Produce distribution plots, time-series plots, and box plots for PM2.5 (and other relevant fields). Pay particular attention to outliers -flag them, and keep track of how you treat them.
- Build **calendar plot(s)** of the unscaled, cleaned PM2.5 data, colour-coded using the AQI categories from Section 4, so that air quality patterns across the six-month window are visible at a glance.
- Build **time-series plots** with shaded/coloured bands across the chart representing the AQI categories from Section 4.
- Apply a **Z-score transformation** to the PM2.5 values for all sensors, and interpret what the scaled values mean.
- For the same ~5 Kampala sensors, examine the **correlation between temperature and humidity** and PM2.5.

7.3 Machine Learning Model Implementation

The target variable is **PM2.5** (using the Z-score-transformed values from Section 7.2).

- Train **5 machine learning models** of your choice, with hyperparameter tuning on each. For every model, explain *which* hyperparameters you tuned and *why* those particular ones matter for that model.
- Try at least one **ensemble learning** approach as well, again with hyperparameter tuning where appropriate.
- Choose evaluation metrics deliberately -explain which metric(s) you used and why they are appropriate for this regression task (and when a different metric would be more appropriate).
- Identify the **best-performing model** among the five, and **save it** (serialized model artifact).
- Using the saved model:
 - Predict **daily PM2.5 concentrations** across all sensors for the first two full weeks of **August 2026**.
 - Predict **hourly PM2.5 concentrations** for the first full day of **August 2026**.
 - For both, **inverse-transform** the predictions back from Z-scores to real PM2.5 concentration units ($\mu\text{g}/\text{m}^3$).
 - Visualize the daily predictions on a **calendar plot** and a **time-series plot**, embedding the AQI legend from Section 4 so it is clear which periods are safe versus unsafe (select ~5 sensors of your choice, balanced across zones).

- *Optional (extra credit):* using shapefiles, plot on a map the locations of sensors whose predicted PM2.5 falls in the Unhealthy for Sensitive Groups / Very Unhealthy range by **13:00 on 1st August 2026**.

7.4 Model Monitoring with Evidently AI

Integrate **Evidently AI** and use it to assess **data drift** and **model drift** on your pipeline. Explain, in your report, what each metric/parameter Evidently reports actually means and what it tells you about this specific problem.

7.5 Deployment on RunPod

RunPod is a pay-as-you-go cloud platform for AI/ML workloads. Budget **no more than \$5 USD** for this stage.

- Select a cost-friendly managed service option covering GPU, vCPU, storage, and endpoint usage.
- Choose a **Pod** (a dedicated GPU instance, suited to development, testing, and fine-tuning).
- Pick an affordable, available GPU -check the pod summary and pricing summary before committing.
- Choose a Pod Template (e.g. a PyTorch environment) you're comfortable working in.
- Import your data, run your full pipeline on the Pod, and check the resulting metrics and runtime logs.

Before you touch RunPod: make sure your entire notebook/pipeline runs cleanly end-to-end locally first. Debugging on RunPod costs real money against your \$5 budget -get it right at home, then deploy.

8. Deliverables

Submit all of the following:

1. **Presentation (PPT)** summarizing your approach, findings, and results.
2. **Report (LaTeX)** documenting your full methodology, analysis, and results.
3. **Code file(s)** covering the full pipeline (data preparation through deployment).
4. **Saved best model(s)** -your selected ML and/or ensemble model artifact(s).

9. Submission Notes

- If **Evidently AI** throws an error specifically when running on RunPod, comment out that code block rather than letting the pipeline fail -the rest of the pipeline must still run. **The pipeline must not crash – thou shall not crash 😊** .
- Show your working at every stage: unexplained jumps from raw data to final metrics will cost you marks even if the final numbers look right.

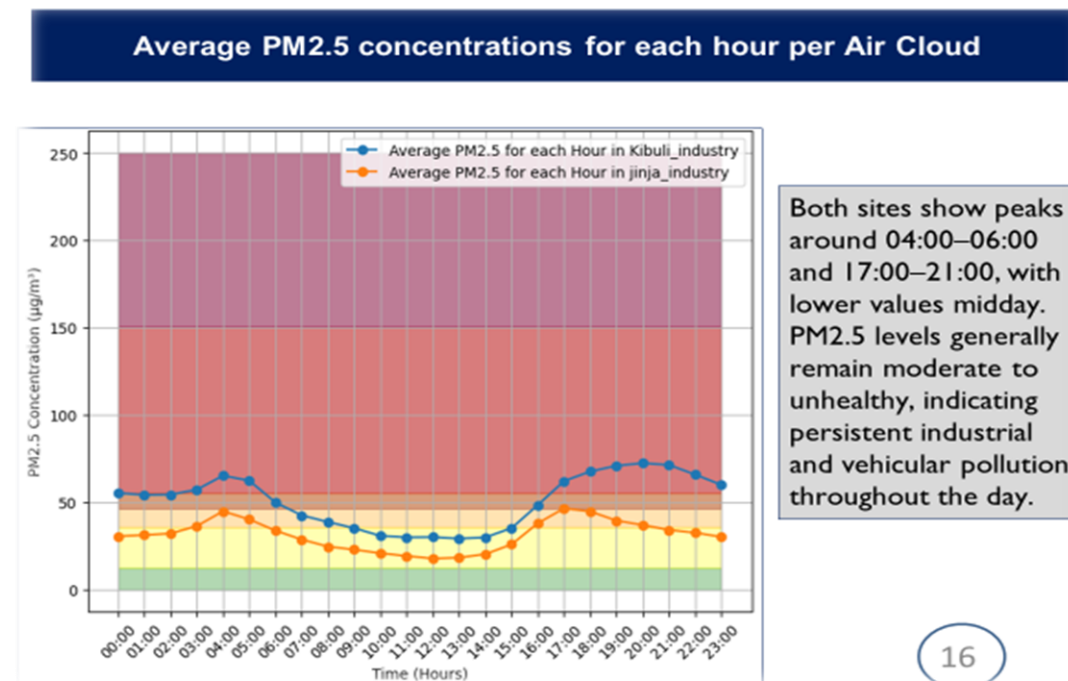
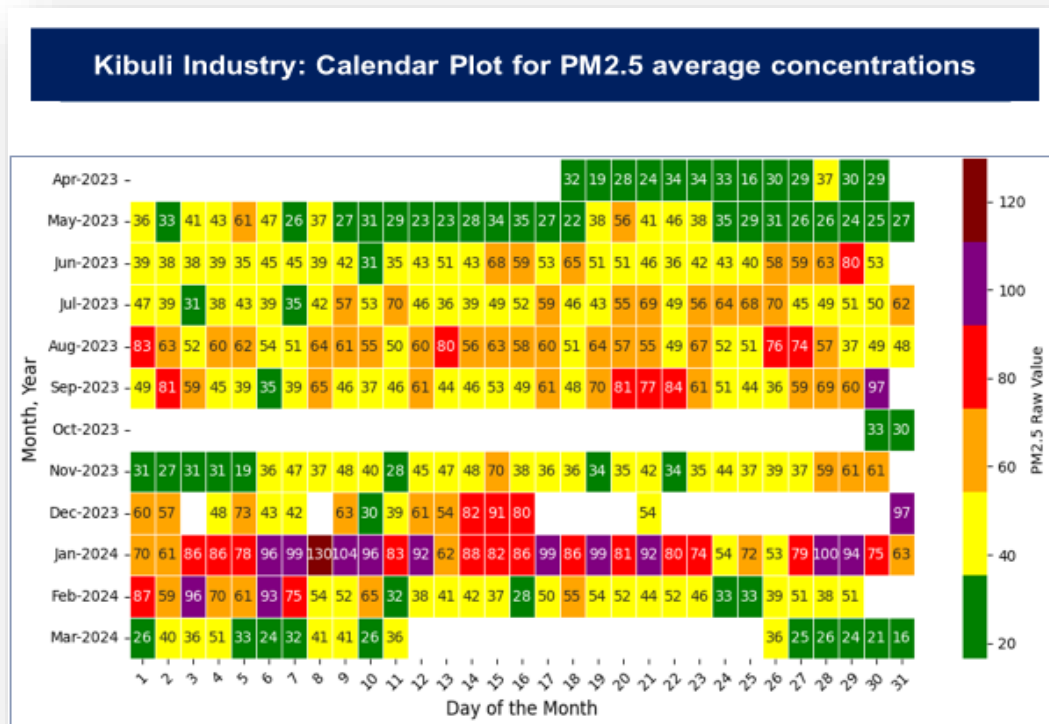
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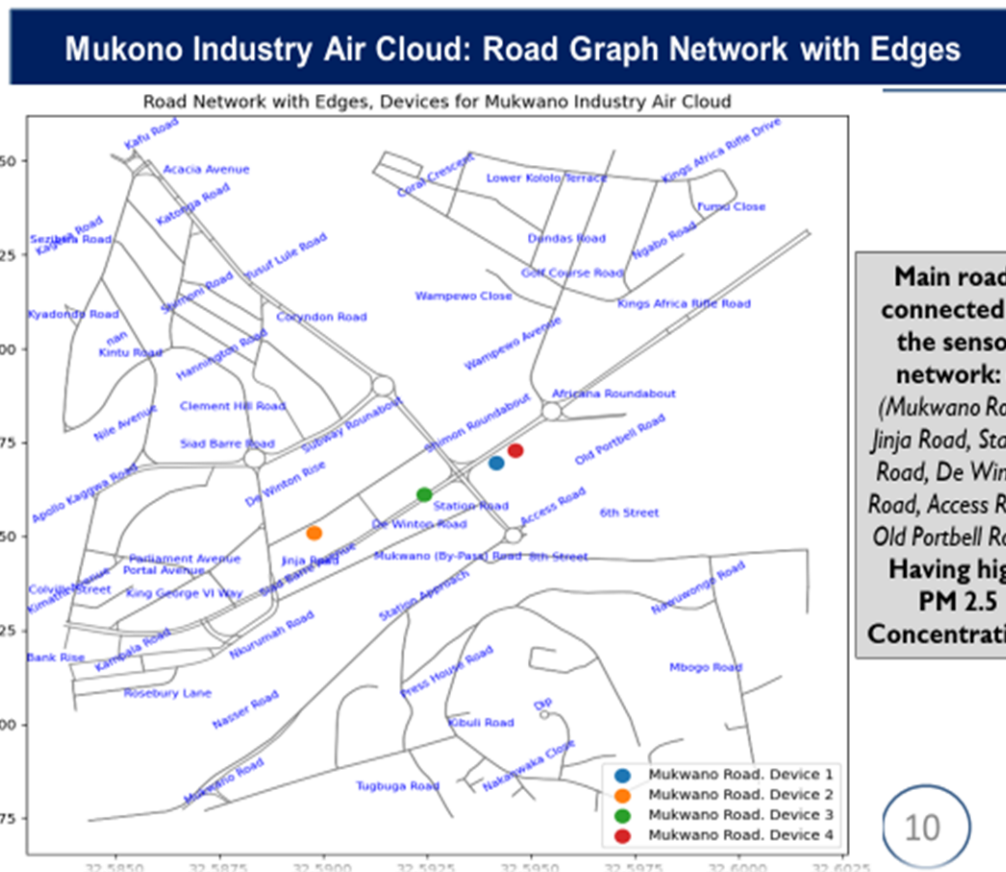
Areas of Assessment.

Here's the assessment summary as bullets: (Draw a methodology pipeline – flow chat)

- **Data preparation** - soundness of null-handling, feature selection reasoning, hourly resampling across Jan–Jul 2026, and data type consistency.
- **Exploratory data analysis** - quality of distribution, time-series, box, and calendar plots; correct use of AQI colour bands; correct Z-score transformation and interpretation; temp/humidity correlation analysis.
- **Model implementation** - breadth and correctness of the 5 ML models and ensemble model(s), justified hyperparameter tuning, appropriate evaluation metric(s), correct identification/saving of the best model.
- **Forecasting and visualization** - correctness of the daily (2-week) and hourly (1-day) August 2026 forecasts, correct inverse Z-score scaling, clarity of calendar/time-series visualizations with the AQI legend.
- **Model monitoring** - depth of understanding shown in the Evidently AI data-drift and model-drift analysis.
- **Deployment** - successful, documented run on RunPod, with metrics and runtime logs captured, and evidence of local debugging before deployment.
- **Deliverable quality and completeness** - all four required outputs (PPT, LaTeX report, code, saved model), clearly written and reproducible.
- (Bonus) -shapefile-based mapping of sensors in the Unhealthy for Sensitive Groups / Very Unhealthy range at 13:00 on 1 August 2026.

Sample Plots.





Run pod architecture cost panel – choose the cheapest at least those below 2USD per hour:

<https://www.runpod.io/>

runpod | Pods > Deploy a Pod > GPU Pod | Ask agent | \$10.04 +

Q Search ctrl+k

Home
Hub
Resources
Serverless
Pods
Clusters
Storage
Account
Templates
Secrets
Audit logs
Create team
Credentials
Feedback
Support & resources

Featured GPUs

GPU Model	Price/hr	VRAM	RAM	vCPU	Max	Quality
H100 SXM	\$3.49/hr	80 GB	125 GB	8 vCPU	7 max	High
RTX PRO 6000	\$2.09/hr	96 GB	140 GB	16 vCPU	7 max	High
H200 SXM	\$4.59/hr	141 GB	188 GB	12 vCPU	8 max	Medium
B200	\$6.79/hr	180 GB	251 GB	24 vCPU	1 max	Low

NVIDIA latest generation

GPU Model	Price/hr	VRAM	RAM	vCPU	Max	Quality
RTX 4000 Ada	\$0.28/hr	20 GB	50 GB	9 vCPU	1 max	Low
RTX 4090	\$0.74/hr	24 GB	41 GB	15 vCPU	1 max	Low
RTX PRO 4000	\$0.57/hr	24 GB	31 GB	12 vCPU	3 max	Low
PRO 6000 MIG 24GB	\$0.59/hr	24 GB	47 GB	4 vCPU	24 max	Low
RTX 5090	\$0.99/hr	32 GB	94 GB	15 vCPU	7 max	Low
RTX PRO 4500	\$0.72/hr	32 GB	62 GB	12 vCPU	3 max	Low
RTX PRO 4500 SE	\$0.72/hr	32 GB	94 GB	16 vCPU	3 max	Low
L40S	\$1.09/hr	48 GB	94 GB	16 vCPU	6 max	Low
PRO 6000 MIG 48GB	\$1.09/hr	48 GB	94 GB	8 vCPU	11 max	Low
H100 PCIe	\$2.89/hr	80 GB	251 GB	16 vCPU	1 max	Low
H100 NVL	\$3.19/hr	94 GB	141 GB	16 vCPU	5 max	Low
RTX PRO 6000 WK	\$2.19/hr	96 GB			8 max	Unavailable